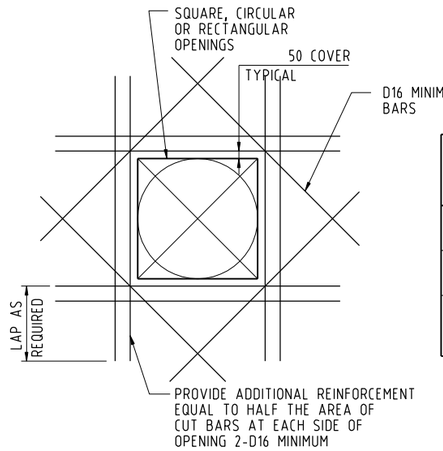
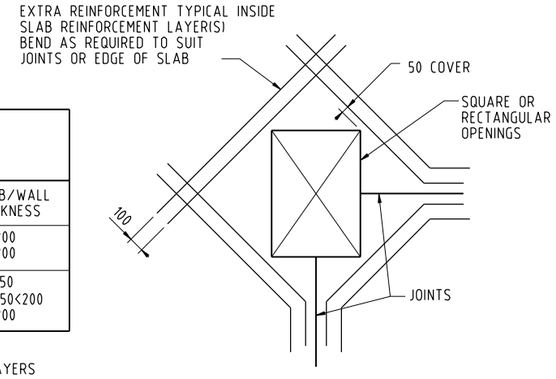
MA'ADEN PHOSPHATE PROJECT-A1 BORDER-REV A02



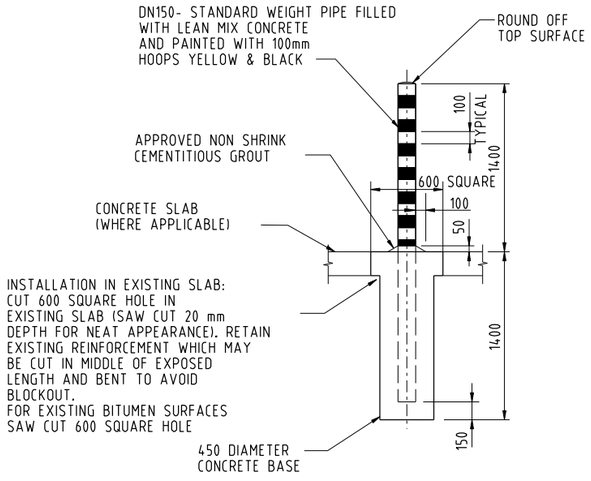
RE-ENTRANT BARS AT SLAB OR WALL OPENINGS			
OPENINGS	MINIMUM EXTRA REINFORCEMENT	LENGTH OF BAR	SLAB/WALL THICKNESS
CIRCULAR	2-D20 2-D24	- -	<200 >200
SQUARE OR RECTANGULAR	2-D20 2-D24 2-D28	1200 1600 2000	<150 >150<200 >200

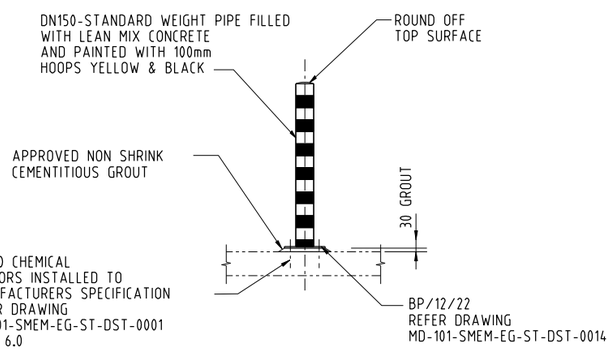
RE-ENTRANT BARS ARE REQUIRED AT ALL LAYERS OF REINFORCEMENT IF MORE THAN ONE LAYER

DETAIL 1/- REINFORCEMENT AT SLAB OR WALL OPENINGS (WITHOUT JOINTS)



DETAIL 2/- REINFORCEMENT AT SLAB OR WALL OPENINGS (WITH JOINTS)
ADDITIONAL SIDE BAR REINFORCEMENT NOT SHOWN FOR CLARITY REFER NOTE 2





ALTERNATIVE DETAIL AT EXISTING SLAB

NOTE:
DEPTH OF EXISTING SLAB TO BE CHECKED PRIOR TO SELECTION OF CHEMICAL ANCHORS.

ELEVATION - TRAFFIC BOLLARD (TYPICAL)

DETAIL 3/-

NOTES:

1. FOR CONCRETE GENERAL NOTES REFER DRG MD-101-SMEM-EG-ST-DST-0001
2. FOR DETAILS OF REINFORCEMENT REFER DESIGN DRAWINGS

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	MD

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REV C01 PREPARED FOR MA'ADEN BY HATCH ASSOCIATES H332701-0000-35-041-0007_C

MA'ADEN معادن Saudi Arabian Mining Company

DRAWN	J BADAL	01.03.11
CHECKED	J EDWARDS	31.03.11
ENGINEER	D KIRKWOOD	31.03.11
APPROVED	R COVELLA	11.05.11
MA'ADEN		
SCALE NTS		A1

MA'ADEN ENGINEERING MANUAL
ALL AREAS
STRUCTURAL STANDARDS
BOLLARD AND RE-ENTRANT BAR DETAILS

DRAWING NUMBER MD-101-SMEM-EG-ST-DST-0007 REV C01

$FILE$ $TIME$ $DATE$